

B. Tech. (3rd Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security, CSE with specialization in Big Data and Analytics, CSE with specialization in Full Stack Development, CSE with specialization in AI & Machine Learning)

BCSE-503 (Database Management System)

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Continuous evaluation	40
End semester exam	60
Total marks	100
Credits	3.0

Course Objectives:

- To understand the concept of DBMS and RDBMS.
- To study the physical and logical database designs & schema.
- To understand DDL, DQL, DML and DCL to create, query, update, and manage a database.
- To impart knowledge about the concept of Normalization to design & build a database system for different applications.
- To impart knowledge about concurrency control, recovery techniques and database security of database systems.

Unit:-1**(9 Hours)**

Database system architecture- Data Abstraction, Data Independence, Various views of data, data models, Schemes, Logical database design, Database design issues, Responsibility of database administrator, Three levels architecture of database systems, Introduction to Client/Server architecture. Data Definition Language (DDL), Data Manipulation Language (DML). Data models- Entity-relationship model, network model, relational and object oriented data models, integrity constraints, data manipulation operations.

Unit:-2**(9 Hours)**

Relational query languages- Relational algebra, Tuple and domain relational calculus, SQL3, DDL and DML constructs, Open source and Commercial DBMS - MYSQL, ORACLE, DB2, SQL server. Relational database design- Domain and data dependency, Armstrong's axioms, Normal forms, Dependency preservation, Lossless design. Query processing and optimization- Evaluation of relational algebra expressions, Query equivalence, Join strategies, Query optimization algorithms.

Unit:-3**(9 Hours)**

Storage strategies- Indices, B-trees, hashing. Introduction to Query Languages- QBE, integrity constraints, functional dependencies & normalization (Normal forms – up to 5th Normal Form). Transaction processing- Concurrency control, ACID property, Serializability of scheduling, Locking and timestamp based schedulers, Multi-version and optimistic Concurrency Control schemes, Database recovery.

Unit:-4**(9 Hours)**

Database Security- Authentication, Authorization and access control, DAC, MAC and RBAC models, Intrusion detection, SQL injection. Advanced topics- Object oriented and object relational databases, Logical databases, Web databases, Distributed databases, Data warehousing and data mining.

Course Outcomes: After completion of this course, student should be able to:

- Understand the concept of DBMS and RDBMS.
- Acquire knowledge about the physical and logical database designs & schema.
- Understand DDL, DQL, DML and DCL to create, query, update, and manage a database.
- Impart knowledge about the concept of Normalization to design and build a database system for different applications.
- Explain about concurrency control, recovery techniques and database security of database systems.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

- Abraham Silberschatz, Henry F. Korth, S. Sudarshan , "Database System Concepts", 6th Edition, McGraw-Hill.
- J. D. Ullman, "Principles of Database and Knowledge – Base Systems", Vol 1, Computer Science Press.

Reference Books:

- R. Elmasri and S. Navathe, "Fundamentals of Database Systems", 5th Edition, Pearson Education.

2. Serge Abiteboul, Richard Hull, Victor Vianu, "Foundations of Databases", Reprint, Addison-Wesley.